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Suspension assembly

Abstract

A suspension assembly for suspending a lower structure such as a platform or platform portion from an upper structure such as ad bridge. A sling includes a cable one end of which is attachable to the upper structure and an other end of which has an eyelet formed by folding an end portion of the cable back over the cable and swaging the end portion of the cable to the cable. A chain having a plurality of links and has an end portion which is attached to the eyelet by receiving the end portion of the cable in one of the links in the end portion of the chain before the end portion of the cable is swaged to the cable. The chain has an other end portion which is attachable to the lower structure by a selected one of its links to be adjustably suspended from the upper structure.

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Background/Summary

(1) This application is a divisional of application Ser. No. 17/229,791, filed Apr. 13, 2021, which is a divisional of application Ser. No. 16/812,129, filed Mar. 6, 2020, which applications are hereby incorporated herein by reference.

(1) The present invention relates generally to scaffolding, for example, the temporary erection of platforms below a bridge deck so that cleaning, painting, or other maintenance work may be performed thereon. More particularly, the present invention relates to suspension devices such as

slings used, for example, for attaching such platforms or other scaffolding members to overhead structures such as bridge decks so that they are suspended at a desired height therefrom. The present invention is also applicable for the suspension of other things.

(2) Applicant's company, Safespan Platform Systems, Inc., has for many years provided and erected temporary platforms or scaffolding below bridge decks, as exemplified in U.S. Pat. Nos. 5,730,248; 5,921,346; 6,003,634; 6,135,240; 6,138,793; 6,227,331; 6,264,002; 6,302,237; 6,386,319; 6,523,644; 8,123,001; 9,217,451; 9,309,633; and 9,784,001; 9,896,852; 10,266,998; 10,267,349; and 10,280,635, and U.S. published application 2018/0135316, all of which patents and published application are incorporated herein by reference.

(3) Such a sling is illustrated at 32 in FIGS. 1, 2, and 30 of the aforesaid U.S. Pat. No. 6,523,644 (wherein it is called an auxiliary support cable), wherein the lower ends of slings are attached to a platform (which is also supported by underlying cables) and their upper ends are attached to bridge structure.

(4) Such a sling is also illustrated at **12** in FIG. **1** of the drawings for the present application. This sling **12** is of a type which is non-adjustable and has been used for many years in scaffolding with great effectiveness when the platform height or other height is fixed and requires no adjustability. This sling **12** will be discussed hereafter in the Detailed Description of the Preferred Embodiment(s).

(5) There are often instances requiring the adjustment of the lengths of the slings in-situ or in the field during the erection process, and length adjustable slings, wherein the eyelet at one end is formed in the field during the erection process, have been provided for this purpose. A length-adjustable sling is illustrated generally in FIG. 3 of the aforesaid U.S. Pat. No. 9,784,001, and includes a permanent swaged connection at one end. The other end of the cable is looped around to also form an eyelet. Instead of a swage, cable portions are held together to form this eyelet by a plurality of suitable clamps which can be loosened to adjust the sling length then tightened at that desired adjusted sling length. The clamps are typically applied in the field during the erection process, then the cable end pulled through until the desired height or sling length is obtained, then the clamps tightened. While such adjustable slings are effective for their purpose, it undesirably takes a long time to put on the clamps and conduct the adjustment. Moreover, the tightening of the clamps undesirably puts kinks in the cables with the result that it is considered unsatisfactory to re-use the slings. Thus, there has been a long-existing need in the scaffolding industry for suspension assemblies which include slings wherein such a suspension assembly can be more easily length-adjusted and which are satisfactory for re-use.

(6) Chains have been provided with hooks at their ends, and the length has been adjusted (to achieve the desired adjusted height of a suspended structure) by looping (or choking) an end portion of a chain around and attaching it back to itself with a grab hook, such as illustrated in FIG. 6 of U.S. Pat. No. 4,854,419, which is incorporated herein by reference. The chains are undesirably too heavy in longer lengths.

(7) Devices have also been provided which comprise a cable and a chain the ends of which are attached together with a shackle or other piece of hardware. Such shackles (or other hardware) as well as hooks used on the other ends of the chain are undesirably expensive.

(8) Applicant's aforesaid U.S. Pat. No. 9,784,001 discloses a sling attached to an adjustment device for suspending one structure such as a platform at a desired distance below another structure such as a bridge structure. The adjustment device comprises a pair of elongate spaced apart plates which are attached to each other and to a hook at one end, the hook attachable to the one structure. The plates have a plurality of longitudinally spaced aligned holes respectively for receiving a pin for connecting one end of the sling to the adjustment device at a selected incremental one of the pairs of aligned holes. The other end of the sling is connected to the another structure to thereby adjust the distance over which the one structure is suspended below the another structure. While such an assembly is a very effective enhancement, the hook is nevertheless expensive.

- (9) It is accordingly an object of the present invention to provide an alternative suspension assembly wherein the length or height over which it is used can be easily adjusted and without the use of expensive hooks or other hardware.
- (10) In order to provide such an alternative suspension assembly without the use of expensive hooks, in accordance with the present invention, an adjustment assembly is provided wherein a chain is attached to a sling, and adjustment is provided by cinching a chain link in a less expensive none-hook device which is easily and inexpensively attachable to a platform or other structure to be suspended.
- (11) In order to eliminate an expensive shackle or other hardware for attaching ends of the sling and chain, in accordance with the present invention, the sling is a cable an end portion of which is received in a chain link and looped around and swaged to itself, to thereby provide an inexpensive connection of the chain to the sling.
- (12) The above and other objects, features, and advantages of the present invention will be apparent in the following detailed description of the preferred embodiment(s) thereof when read in conjunction with the accompanying drawings wherein the same reference numerals denote the same or similar parts.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a side view, partly schematic, of an adjustable suspension assembly in accordance with the present invention, illustrated suspending a platform from a bridge structure.
- (2) FIGS. 2 and 3 are enlarged perspective view of a portion of the assembly.
- (3) FIG. 4 is a blown up view of a portion of the assembly.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT(S)

- (4) Referring to the drawings, there is shown generally at **10** a suspension assembly, which comprises a combination of a conventional sling **12**, a chain **14** one end **16** of which is attached to an end **18** of the sling **12** and an other end of which is attached to an adjustment device, illustrated generally at **20**, in a manner as described hereinafter for providing adjustability to the overall length or height, illustrated at **22**, of the suspension assembly **10** for thereby providing a desired height to a platform **24** to which the adjustment device **20** is attached. Each of the parts thereof, unless otherwise specified or apparent, is composed of steel or other suitable metal or other suitable material.
- (5) The sling **12** is a length of cable **15** (or wire rope or other suitable flexible strand) having attachment means in the form of a loop or eye or eyelet **19** at each end wherein the cable **15** is folded back over and attached to itself to form an eyelet or loop, and a protective thimble (not shown) suitably received within the eyelet **19**. The sling **12** shown in FIG. 1 is of a type which is non-adjustable, wherein, for each eyelet **19**, the respective end portion of the cable **15** is looped back and attached to itself permanently by a swaged connection utilizing a swage sleeve **17** (which may come in various sizes such as, for example, a diameter of ½ inch) which is caused to encircle and firmly grip the respective portions of the cable **15** thereby providing a strong connection of the cable to itself thereby forming the eyelet **19**. Such a swaging process for forming an eyelet of a sling is well known in the art to which the present invention pertains. The upper eyelet **19** (the eyelet opposite end **18**) is connected to a bridge or other suitable structure, illustrated at **26**, by a suitable attachment means such as a clip or hook, illustrated at **28**, for use of the suspension assembly **10** for supporting, for example, the platform **24** from the bridge structure **26**.
- (6) The chain **14** is formed of a plurality of, for example, 21 to 25 interconnected identical links **30** (as needed to provide the desired range of height adjustment), the upper end portion or link **16** (preferably the end link) of which is received on the lower end or eyelet **18** prior to the above

described swaging process thereby desirably permanently and inexpensively connecting the upper end **16** of the chain to the lower end **18** of the sling **12**.

(7) Unless otherwise stated, illustrative dimensions provided herein are for exemplary purposes only and not for purposes of limitation. Each link **30** for an exemplary chain has, for example, a thickness, a width, and a height, illustrated at **32**, **34**, and **36** respectively, which, for a suitable chain **14**, may, for example, be $\frac{3}{8}$ inch, $1\frac{1}{4}$ inch, and 2 inches respectively, the width **34** being substantially greater than the thickness **32**. The chain **14** is of a type wherein, as viewed from a particular direction such as in FIG. 1, alternately the thicknesses **32** and the widths **34** face in that particular direction and have very small or limited rotation ability therefrom. Thus, as viewed in FIG. 1 of the drawings, alternately the thicknesses **32** and the widths **34** face the viewer.

(8) Referring to FIGS. 2 to 4, the device **20** comprises a plate **40** bent or otherwise suitably formed (for example, three flat plates may be welded together to form the plate **40**) to define a bottom wall **42** and two side walls **44** which extend upwardly from edges of the bottom wall **42**. The spacing, illustrated at **46**, between the side walls **44** is greater than the width **34** of a link to allow passage easily of the width **34** of a link there between, as illustrated in FIG. 2. This spacing **46** may, for example, be about $1\frac{1}{2}$ inch.

(9) Each of the side walls **44** has in its upper portion a pair of spaced apertures, illustrated at **48**, wherein the apertures **48** in one of the side walls **44** are aligned with the apertures **48** respectively in the other of the side walls **44**. A pin **52** is insertable in each aperture **48** in one side wall **44** and the corresponding aligned aperture **48** in the other side wall **44**. The pins **52**, which have heads **54**, are desirably inserted from opposite side walls **44** respectively, as illustrated in FIG. 2. The apertures **48** in each side wall **44** are spaced a distance, illustrated at **50**, so that when the pins **52** are inserted in respectively aligned apertures **48**, the distance between the pins **52** will be equal substantially or a little more than the thickness **32** of a link and will be substantially less than the width **34** of a link. This distance **50** may, for example, be about $\frac{5}{16}$ inch. While this exemplary distance **50** is less than the exemplary link thickness of $\frac{3}{8}$ inch, this will nevertheless work because the pin diameters, illustrated at **49**, are substantially less than the diameters, illustrated at **47**, of apertures **48** thereby to provide plenty of play and wiggle room for receiving a link thickness between the pins and allows the distance between the pins **52** to spread to a distance, for example, 1 inch, which is greater than the distance **50** between the apertures and greater than the exemplary $\frac{3}{8}$ inch link thickness for easily accommodating a link thickness.

(10) In order to retain the pins **52** in the respective aligned apertures **48** without the necessity of applying nuts, the pins **52** are preferably scaffold pins, which may also be referred to as adjustment retainers and which are disclosed in the aforesaid U.S. Pat. No. 9,784,001. The scaffold pin includes, in addition to the shank **56** and head **54**, a locking part **58** which utilizes gravity for retaining the shank **56** in the respective aligned apertures **48** as follows. The locking part **58** includes a first portion **60** which extends from the head **54** downwardly, a second portion **62** which extends from an end of the first portion **60** along side a lower portion of the respective side wall **44** to just beyond the nearest edge of the respective side wall **44**, a third portion **64** which extends from the end of the second portion **62** across both side walls **44**, and a fourth portion **66** which extends from the end of the third portion **64** along the respective side wall **44** to a point **68** of termination. It can be seen in FIG. 2 that the fourth portions **66** restrain the pins **52** from being removed from their apertures **48** (due to impingement of the fourth portions **66** against the side walls **44** respectively). It can also be seen that, by movement of the locking parts **58** upwardly, the fourth portions **66** can be made to clear the side walls **44** respectively so that the pins **52** can then be removed from the apertures **48** respectively. The side edges of the side walls **44** are slightly curved inwardly, as seen at **57** in FIG. 2, to allow ease of movement of the locking parts **58** between the lower and upper positions. The force of gravity keeps the locking parts **58** in their downward positions as seen in FIG. 2. When force is applied to the suspension assembly **10** such as by attachment of the platform **24**, the pins **52** are pinched in the respective apertures **48** thus further

insuring that the pins **52** will not come out of their apertures **48** respectively.

(11) Additional apertures **61** (or at least one additional aperture **61**) are provided to the sides of apertures **48** respectively to allow attachment of a lifting device for temporary lifting of the platform **24** so that the cable **14** may be attached to or detached from the device **20** or for any other suitable purpose.

(12) The platform **24** is supported by cables **63** such as shown in, for example, the aforesaid U.S. Pat. No. 5,730,248. The decking **24** is attached to the cables **63** at multiple locations by clips **65** comprising a pair of plates **67** which abut (end edge to end edge) to cover openings (not shown) in the decking **24** and a U-bolt **71** which receives a cable **63**, and nuts **70** applied to its two threaded end portions **72** thereby securing the cable **63** to the decking **24** at that point. Similar clips are also shown in the aforesaid U.S. Pat. No. 5,730,248 and in others of the aforesaid patents. For example, a clip may have a J-bolt with one end connected to a plate and the other end receiving a nut. The use of underlying cables to support decking for platforms and the attachment thereof to the decking by clips of various types is well known in the art.

(13) In accordance with the present invention, the end portions **72** of the U-bolt are also received in apertures, illustrated at **74** (one shown), in the bottom wall **42** of the device **20** before the nuts **70** are applied, thereby attaching the device **20** to the platform **24**, without the need for expensive hooks or the like. If a different type of clip is used, it may suitably attach the platform **24** to the device **20** using principles commonly known to those of ordinary skill in the art to which the present invention pertains.

(14) In order to suspend the platform **24** at the desired height, the sling eyelet **19** (the one unattached to the chain **14**) is suitably attached to an overhanging bridge portion **26** such as by receiving the eyelet on the hook **28**. The device **20** as well as an underlying cable **63** are attached to the platform with the U-bolt **71** or as is otherwise suitable. Using a temporary suspension device such as another sling attached at one of the apertures **61** to maintain the platform **24** temporarily lifted, the correct chain link **30** for the desired platform height is selected and its thickness **32** inserted in position between the side walls **44** and the pins **52** inserted into the respective aligned apertures **48** (while suitably holding the locking parts **58** above the side walls **44**, then allowing the locking parts to fall by gravity and urging as necessary into locking position after the pins **52** are inserted), thereby cinching the selected link **30** for the desired platform height into position. As seen in FIG. 2, the adjacent links **30** cannot pass width-wise through the spacing **50** between the pins **52** thereby trapping or locking the selected link **30** at the desired height **22** and thereby providing an adjusted height to the suspension assembly **10**. To remove the suspension assembly **10**, the temporary suspension device is again applied at one of the apertures **61** to relieve force, the locking parts **59** are moved so that the fourth portions **66** are above the side walls **44**, and the pins **52** are removed from the apertures **48** respectively and the chain **14** removed from the device **20**.

(15) It should be understood that the clip **65** as well as underlying cables **63** are not essential to the present invention and that the suspension assembly **10** may be used to suspend things other than platforms by suitable attachment of the device **20** thereto.

(16) It should be understood that, while the present invention has been described in detail herein, the invention can be embodied otherwise without departing from the principles thereof, and that such other embodiments are meant to come within the scope of the present invention as defined by the appended claims.

Claims

1. A suspension assembly comprising: a sling including a cable one end of which is attachable to a first structure and an other end of which has an eyelet formed by folding an end portion of said cable back over said cable and swaging said end portion of said cable to said cable; a chain having a series of successively interconnected links which are shaped in the form of rings, wherein each of

said links passes through adjacent one or ones of said links, said chain further having an end portion which has at least one of said links and which is attached to said eyelet by receiving said end portion of said cable in said at least one of said links in said end portion of said chain before said end portion of said cable is swaged to said cable, and said chain has an other end portion which is attachable to a second structure for suspension of said second structure from the first structure, wherein said other end portion of said chain has at least two of said links whereby the height of the second structure is adjustable by attachment of one or the other of said at least two of said links to the second structure.

2. An assembly according to claim 1 wherein the first structure is a bridge and the second structure is a platform.

3. An assembly according to claim 1 wherein said series of links comprises at least twenty-one of said links.

4. A suspension assembly comprising: a sling including a cable one end of which is attachable to a first structure and an other end of which has an eyelet formed by folding an end portion of said cable back over said cable and swaging said end portion of said cable to said cable; and a chain having a plurality of links and having an end portion which is attached to said eyelet by receiving said end portion of said cable in one of said links in said end portion of said chain before said end portion of said cable is swaged to said cable, and said chain has an other end portion which is attachable to a second structure for suspension of said second structure from the first structure, the assembly further comprising a device for attaching a selected one of said links of said other end portion of said chain to the second structure, wherein said device includes a first plate portion attachable to the second structure, a pair of generally parallel second plate portions, and means for cinching said selected one of said links between said second plate portions to adjust a suspended height of the second structure, said means for cinching including first and second pins, a first pair of aligned apertures in said second plate portions respectively for receiving said first pin, and a second pair of aligned apertures in said second plate portions respectively for receiving said second pin, wherein said first and second pins, when received in said first and second pairs respectively of aligned apertures, are spaced apart a distance which allows passage of a thickness of each of said links and which restrains passage of a width of each of said links, whereby, when said selected one of said links is received between said pins while said pins are received in said first and second pairs respectively of aligned apertures, ones of said links which are adjacent said selected one of said links are restrained from passage between said pins.

5. An assembly according to claim 4 wherein at least one of said second plate portions has at least one additional aperture for attachment of a lifting device to provide temporary lifting of the second structure while said selected link is being cinched.

6. An assembly according to claim 4 wherein said pins are scaffold pins.

7. An assembly according to claim 4 wherein the second structure is a platform having underlying cables, the assembly further comprising a clip including a curved shank having at least one threaded portion for attaching the platform to one of the underlying cables, at least one aperture in said first plate portion for receiving said at least one threaded portion of the shank, wherein said one of the underlying cables is attachable to the platform with the clip, and said at least one threaded portion is passed through said at least one aperture in said first plate portion, and a nut is applied to said at least one threaded portion for attaching said device to said clip.

8. An assembly comprising: a first structure; a second structure disposed below said first structure; a sling including a cable one end of which is attached to said first structure and an other end of which has an eyelet formed by folding an end portion of said cable back over said cable and swaging said end portion of said cable to said cable; a chain having a series of successively interconnected links which are shaped in the form of rings, wherein each of said links passes through adjacent one or ones of said links, said chain further having a first end portion which has at least one of said links and is attached to said eyelet by receiving said end portion of said cable in

said at least one of said links in said first end portion of said chain before said end portion of said cable is swaged to said cable, and said chain has a second end portion which is attached to said second structure thereby suspending said second structure from said first structure, wherein said other end portion of said chain has at least two of said links whereby the height of said second structure is adjustable by attachment of one or the other of said at least two of said links to said second structure.

9. An assembly according to claim 8 further comprising a device attaching a selected one of said links of said second end portion of said chain to said second structure, wherein said device includes a first plate portion attached to said second structure, a pair of generally parallel second plate portions, and means for cinching said selected one of said links between said second plate portions, whereby suspended height of the second structure is adjusted, said means for cinching including first and second pins, a first pair of aligned apertures in said second plate portions respectively for receiving said first pin, and a second pair of aligned apertures in said second plate portions respectively for receiving said second pin, wherein said first and second pins, when received in said first and second pairs respectively of aligned apertures, are spaced apart a distance which allows passage of a thickness of each of said links and which restrains passage of a width of each of said links, whereby, when said selected one of said links is received between said pins while said pins are received in said first and second pairs respectively of aligned apertures, ones of said links which are adjacent said selected one of said links are restrained from passage between said pins.

10. An assembly according to claim 9 wherein at least one of said second plate portions has at least one additional aperture for attachment of a lifting device to provide temporary lifting of said second structure while said selected link is being cinched.

11. An assembly according to claim 9 wherein said pins are scaffold pins.

12. An assembly according to claim 9 wherein said second structure is a platform having underlying cables, the assembly further comprising a clip including a curved shank having at least one threaded portion for attaching said platform to one of said underlying cables, at least one aperture in said first plate portion for receiving said at least one threaded portion of the shank, wherein one of the underlying cables is attached to said platform with the clip, said at least one threaded portion is passed through said at least one aperture in said first plate portion, and a nut is applied to said at least one threaded portion for attaching said device to said clip.

13. An assembly according to claim 12 wherein said first structure is a bridge.

14. An assembly according to claim 8 wherein said first structure is a bridge and said second structure is a platform.

15. An assembly according to claim 8 wherein said series of links comprises at least twenty-one of said links.
